

Multi-Agent System: Theories and Applications

Lecture 10: Synchronization of Nonlinearly Coupled Oscillators – Kuramoto Model

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Outline

- 1 Original Kuramoto Model
- 2 Generalized Kuramoto Models

Model Description

$$\dot{\theta}_i = \omega_i + \frac{K}{n} \sum_{j=1}^n \sin(\theta_j - \theta_i), \quad i = 1, \dots, n.$$

SELF-ENTRAINMENT OF A POPULATION OF COUPLED NON-LINEAR OSCILLATORS

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Temporal organization of matter is a widespread phenomenon over a macroscopic world in far from thermodynamic equilibrium. A previous study on chemical instability¹⁾ implies that a simplest nontrivial model for a temporally organized system may be represented by a macroscopic self-sustained oscillator Q obeying the equation of motion

$$\dot{Q} = (i\omega + \alpha)Q - \beta|Q|^2Q, \quad (1)$$

$\alpha, \beta > 0.$

Consider a population of such oscillators $Q_1, Q_2, \dots, Q_N$ with various frequencies, and introduce interactions between every pair as follows.

$$\dot{Q}_s = (i\omega_s + \alpha)Q_s + \sum_{r=1}^N v_{rs}Q_r - \beta|Q_s|^2Q_s, \quad (2)$$

$r, s = 1, 2, \dots, N.$

We found that it is possible to construct from (2) a soluble model for a community exhibiting mutual synchronization or self-entrainment above a certain threshold value of the coupling strength. Such a type of phase transition has been considered by Winfree²⁾ without resorting to specialized models but only phenomenologically.

Our simplifying assumptions are:

- (I) $v_{rs} = v/N$ independently of r and s ,
- (II) $\alpha, \beta \rightarrow \infty$ but $\alpha/\beta, \omega_s, v = \text{finite}$,
- (III) $N \rightarrow \infty$.

Let us put $Q_s = \rho_s e^{i\varphi_s}$. Owing to the assumption (II), the amplitude ρ_s may be fixed at $\sqrt{\alpha/\beta}$. Thus we have only to consider the equation

$$\dot{\varphi}_s = \omega_s + \frac{v}{N} \sum_r \sin(\varphi_r - \varphi_s). \quad (3)$$

Meaning of Variables

$$\dot{\theta}_i = \omega_i + \frac{K}{n} \sum_{j=1}^n \sin(\theta_j - \theta_i), \quad i = 1, \dots, n$$

K : coupling strength

For oscillators:

- θ_i : phase of oscillator i
- ω_i : natural frequency of oscillator i
- $\dot{\theta}_i$: frequency of oscillator i

For vehicles or robots:

- θ_i : position of vehicle i
- ω_i : initial speed of vehicle i
- $\dot{\theta}_i$: speed of vehicle i

Arising Questions

- (Q1) If synchronization, what is the synchronized frequency?
- (Q2) What are conditions for synchronization?
- (Q3) How do the synchronized states look like?

Answer to (Q1)

Synchronized Frequency

is equal to the average of natural frequencies of oscillators, i.e.,

$$\omega_{sync} = \frac{\omega_1 + \cdots + \omega_n}{n}$$

Answer to (Q2)

Many results exist so far, but

Common philosophy

The coupling strength K should be greater than a critical threshold $K_{critical}$ in order to suppress the difference on natural frequencies ω_i .

A particular result

Sync is achieved if

$$K > K_{critical} \triangleq \max_i \omega_i - \min_i \omega_i$$

and

$$\max_i \theta_i(0) - \min_i \theta_i(0) \leq \gamma \in [0, \pi/2)$$

where

$$\gamma \triangleq \arcsin \frac{K_{critical}}{K}$$

Answer to (Q2)

Mean-field coupling analysis:

- Order parameter:

$$re^{i\psi} \triangleq \frac{1}{n} \sum_{j=1}^n e^{i\theta_j}$$

- Then Kuramoto model becomes:

$$\dot{\theta}_i = \omega_i + Kr \sin(\psi - \theta_i)$$

- The larger r is, the better synchronization is

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Jacobian analysis:

- Denote $f_i(\theta) \triangleq \omega_i + \frac{K}{n} \sum_{j=1}^n \sin(\theta_j - \theta_i)$, $\theta \triangleq [\theta_1 \quad \cdots \quad \theta_n]^T$, $f \triangleq [f_1 \quad \cdots \quad f_n]^T$
- Jacobian matrix:

$$J(f) = -\frac{K}{n} \mathcal{L}(t) \dot{\theta}$$

where $\mathcal{L}(t) \triangleq [\mathcal{L}_{ij}(t)]$, $\mathcal{L}_{ij}(t) \triangleq \cos(\theta_i - \theta_j)$, $i \neq j$, and $\mathcal{L}(t)\mathbf{1}_n = 0 \rightarrow$ consensus form

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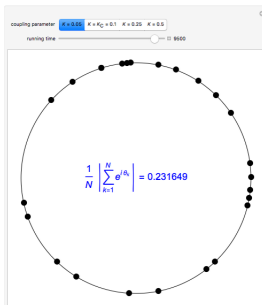
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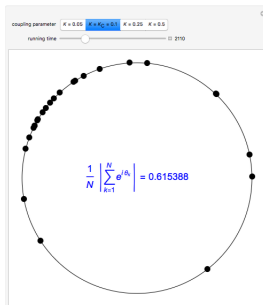
More details and more results can be found in: F. Dörfler & F. Bullo, "Synchronization in complex networks of phase oscillators: A survey", *Automatica*, vol. 50, pp. 1539-1564, 2014.

Answer to (Q3)

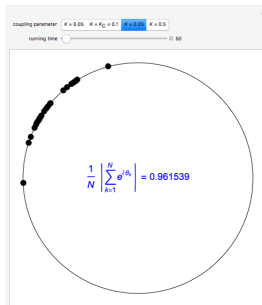
Very rich phenomena on the phases of oscillators



Weak coupling



Critical coupling



Strong coupling

(Source: Wolfram)

A Natural Generalization

$$\dot{\theta}_i = \omega_i + \sum_{j=1}^n a_{ij} f(\theta_j - \theta_i), \quad i = 1, \dots, n, \quad a_{ij} \geq 0.$$

- Sparse structure, not all-to-all like in the original Kuramoto model
- Not equally coupled
- Coupling function f can be any continuous, differentiable, and odd function

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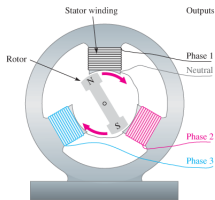
What are conditions for synchronization in this generalized Kuramoto model?

How the synchronized states look like?

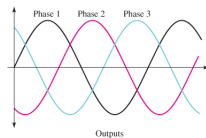
Applications in Electric Power Grids

- Structure-preserving model for power network with synchronous generators:

$$M_i \ddot{\theta}_i + D_i \dot{\theta}_i = P_{m,i} - \sum_{j=1}^n a_{ij} \sin(\theta_i - \theta_j)$$



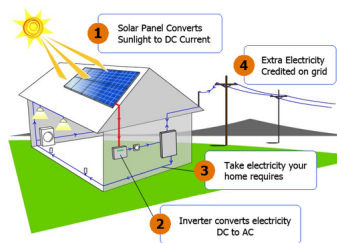
(a) Three-phase rotating-field generator



(b) Three-phase sine wave

- Droop inverter-based microgrid with renewable generation:

$$D_i \dot{\delta}_i + \sum_{j=1}^n a_{ij} \sin(\delta_i - \delta_j) = P_i$$



Other Generalizations

- Disturbances
- Infinite dimension
- Entrainment
- Quantum mechanics related
- etc.